Geospatial Flood-Risk Modelling using Climate Data and Hidden Markov Models in Rwanda

Capstone Defense • Nyabugogo–Nyabarongo corridor, Kigali

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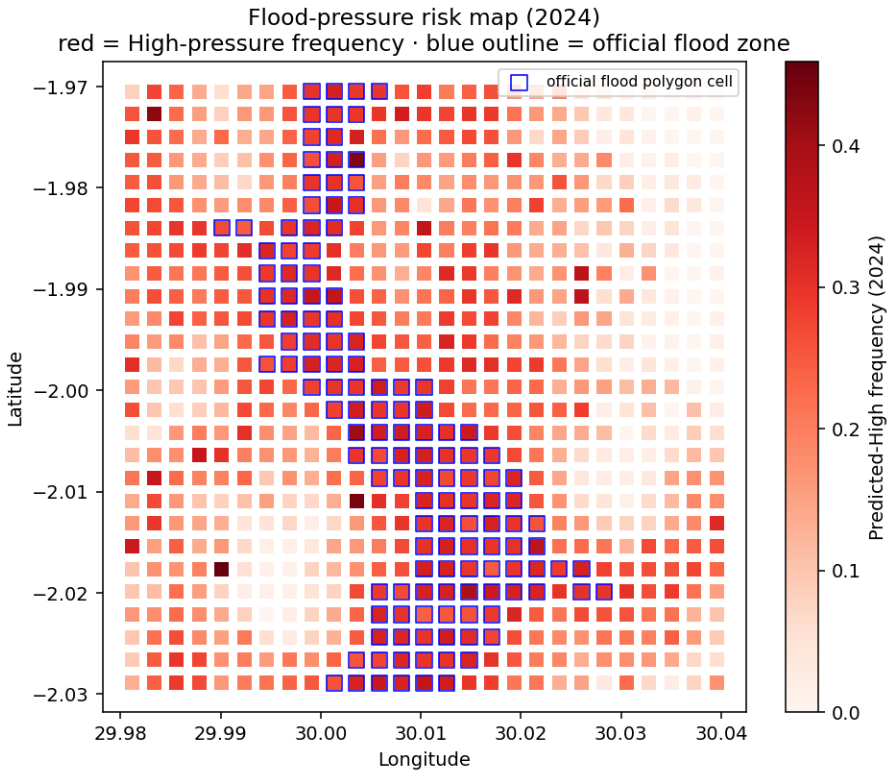
Problem & Objective

- Kigali's catchments flood repeatedly after intense rain; official hazard maps are static.
- They show WHERE flooding is possible — not WHEN or HOW WIDELY pressure rises after rainfall.
- Objective: a geospatial ML + Hidden Markov Model framework that estimates daily
  - Low / Moderate / High flood-pressure states and validates them against official polygons.
- Scope: decision-support / risk-screening — complements, not replaces, early warning.

Study Area & Grid

Nyabugogo–Nyabarongo corridor aligned to the official flood zone

- AOI: lon 29.98–30.04, lat −2.03––1.97 (Kigali).
- 250 m grid → 729 cells.
- Daily, 2018–2024 → 1,864,053 cell-day records.
- Temporal split: train 2018–2022 · validation 2023 · test on unseen 2024.
- AOI deliberately shifted south onto the real MOE flood zone (centroid ≈ 30.054, −1.994).



System Architecture — four layers

- 1. DATA — real, free, keyless feeds per cell-day.
- 2. LABELS — flood-pressure = rainfall trigger × terrain susceptibility (+ unobserved noise).
- 3. MODELS — baselines → RF / XGBoost → HMM temporal layer.
- 4. VALIDATION & DELIVERY — vs official polygons + Streamlit dashboard.
  - Each layer is a separate, reproducible module run by main.py.

Real Data Sources — retrieved live, not assumed

- Rainfall (1/3/7/14-day): Open-Meteo ERA5 archive.
- Elevation + slope: Open-Meteo SRTM.
- Rivers, roads, buildings: OpenStreetMap (Overpass).
- Official flood polygons: Rwanda GeoPortal / geodata.rw (Ministry of Environment).
- EVIDENCE — live ERA5 pull, AOI centroid, Jan 2024:
  - total 131.2 mm over 31 days; e.g. 09 Jan = 12.7 mm, 10 Jan = 4.2 mm.
  - Official layer: 136 / 729 cells (19%) flagged flood-prone.

Features & Labels — why it is not circular

- 10 predictors: 4 rainfall windows + elevation, slope, distance-to-river, road & building density, official-polygon flag.
- Label = top-20% / next-30% / lowest-50% of a rainfall×susceptibility score.
- Added an UNOBSERVED drainage latent + daily noise the model cannot see.
  - Without it, models hit 0.99 F1 — a circular, indefensible result.
  - With it, F1 ≈ 0.81 — realistic and honest.
- Flood-PRESSURE = derived risk state, NOT a confirmed flood event.

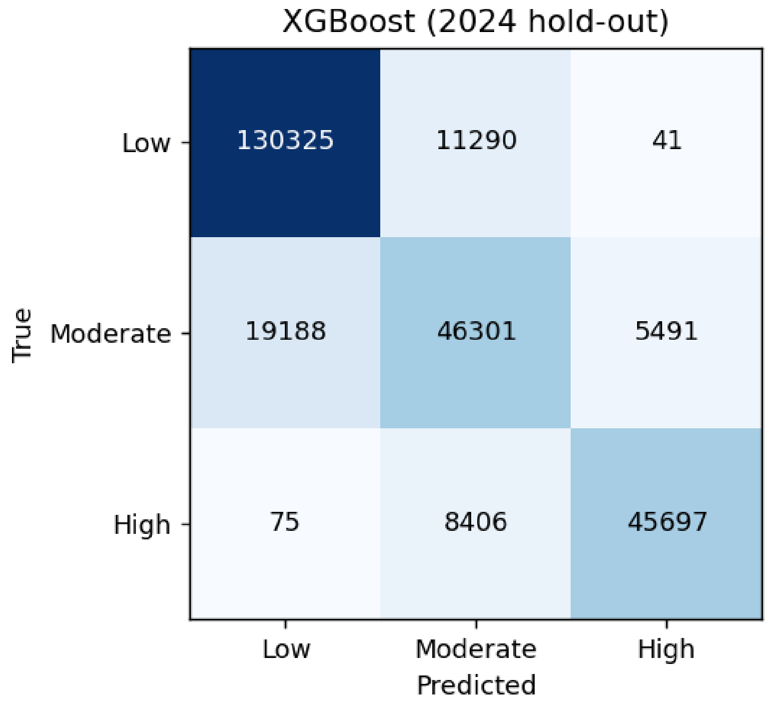
Models & Evaluation Protocol

- Baselines: rainfall-threshold & static-polygon (prove ML adds value).
- Interpretable: Logistic Regression, Decision Tree.
- Main: Random Forest, XGBoost (+ SHAP explainability).
- Temporal layer: Gaussian HMM over daily sequences.
- Temporal split (no shuffle): train 2018–22, validate 2023, test on unseen 2024.
- Train ≈ validation ≈ test macro-F1 (no over-fitting; generalises across years).
- Metrics: macro-F1, High-recall, confusion, next-step accuracy, spatial concordance.

Results — 2024 test hold-out

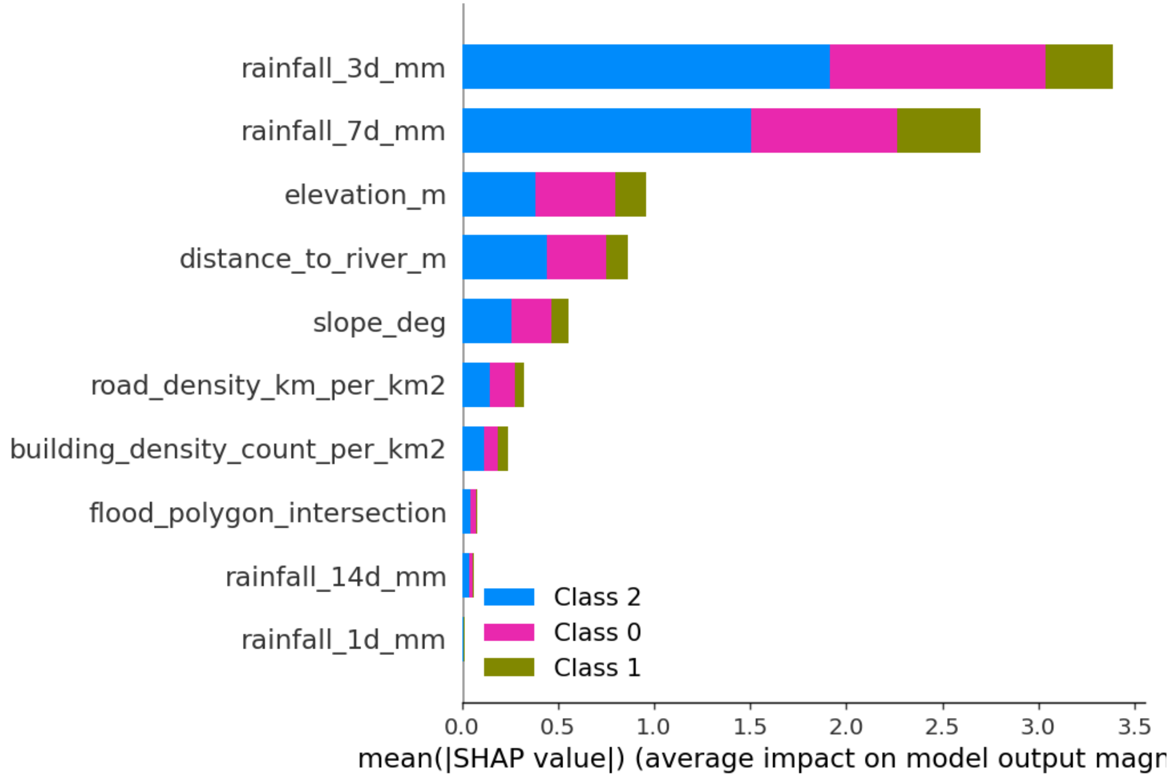
Targets: macro-F1 ≥ 0.75, High-recall ≥ 0.80

Model	Macro-F1	High-recall
XGBoost (deployed, best)	0.813	0.843
Random Forest	0.813	0.849
Decision Tree	0.750	0.798
Logistic Regression	0.750	0.819
Rainfall-threshold (baseline)	0.626	—
Static-polygon (baseline)	0.324	—



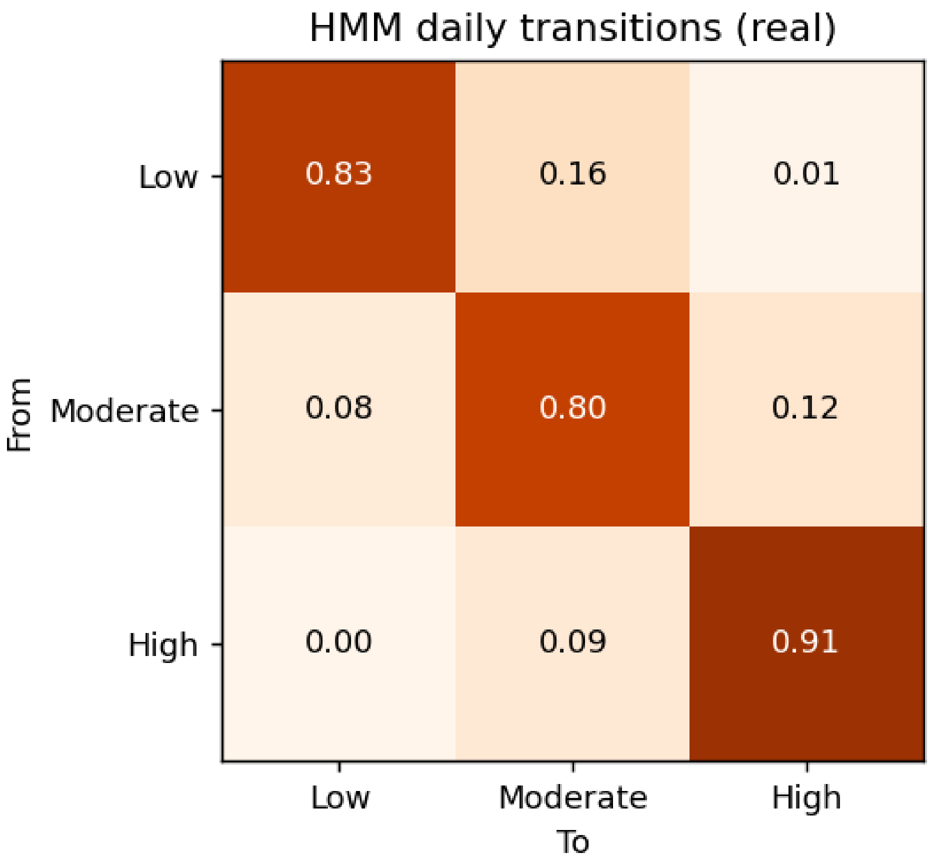
Explainability — SHAP & feature importance

- Rainfall accumulation (3- & 7-day) dominates predicted pressure.
- Then elevation & distance-to-river (low + near water = vulnerable).
- Hydrologically sensible and auditable (satisfies NFR-04).



HMM — temporal dynamics of flood pressure

- Self-transition probabilities ≈ 0.83 / 0.80 / 0.91 (Low/Mod/High).
- Pressure PERSISTS once it sets in after sustained rain.
- Transitions move between neighbouring states, not Low→High jumps.
- Next-step accuracy 0.45 over 3 states (random = 0.33).
- Dashboard now shows a PER-CELL transition matrix: P(state tomorrow | today) for any selected cell — the Markov intuition, cell by cell.



Spatial Validation vs Official Polygons — the honest result

- Official zone covers only 19% of the corridor → a fixed '70% inside' target is geometrically impossible without circular labels.
- Reformulated to enrichment & odds ratio:
  - Raw containment of predicted-High in official zone: 0.30.
  - Enrichment / lift: 1.60× over the 18.7% base rate.
  - Inside-vs-outside High odds: 1.85× (30.7% vs 16.7%).
- Model re-discovers the official zone WITHOUT training on it, and flags a broader dynamic footprint — the project's core contribution.

Delivery — Streamlit Inspection Dashboard

- Pick any date → predicted flood-pressure map over the grid.
- Overlay official flood polygons.
- Inspect any cell: predictors, class probabilities, rainfall time series.
- Per-cell Markov transition matrix + train/validation/test performance panels.
- Model & HMM performance panels built in.
- Run: python main.py dashboard

Rainfall source: ERA5 vs CHIRPS vs Meteo Rwanda

What is feasible — tested, not assumed

• ERA5 (Open-Meteo) — USED. Keyless REST, 1940–now, fully reproducible; verified live (131.2 mm, Jan-2024 AOI). Reanalysis, ~9–25 km.

• CHIRPS — scientifically ideal for Africa (0.05°, satellite+stations). BUT access is heavy: IRI Data Library & ClimateSERV both failed/timed out in testing; needs GeoTIFF/GEE tooling.

• Meteo Rwanda / ENACTS — most authoritative (national gauges) but access is restricted/registration-gated; no open API.

• Recommendation: ERA5 as primary (reproducible now); CHIRPS as a documented validation extension; Meteo Rwanda via formal data request.

Limitations & Future Work

- Labels are derived flood-pressure, not confirmed events (no open incident feed).
- ERA5 is coarser than a dense gauge network.
- Official national polygons are low-resolution.
- Future: MINEMA incident validation; CHIRPS/Meteo Rwanda blend; TWI & HAND features; Optuna tuning; operational daily dashboard.

Conclusion

- All five objectives met on REAL, open data with a temporal train/val/test split.
- XGBoost (deployed) macro-F1 0.813, High-recall 0.843; baselines clearly beaten.
- Model independently reinforces (1.60×) and extends the official flood map.
- A reproducible, interpretable, time-aware flood-pressure tool for Kigali.

Anticipated Panel Questions

- “Is your label circular?” → No — unobserved drainage latent + noise; F1 0.81 not 0.99.
- “Why ERA5 not CHIRPS?” → Reproducible keyless access; CHIRPS tested but fragile (future work).
- “Why did spatial agreement miss 70%?” → Coarse 19% official layer; reframed to enrichment 1.60×.
- “Pressure vs event?” → Derived risk state; event validation needs MINEMA records.
- “Does it generalise?” → Tested on a future unseen year (2024), not random split.